A. All Lengths Subtraction

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, two pointers | https://codeforces.com/problemset/problem/2143/A

You are given a permutation[□] p of length n.

You must perform exactly one operation for each integer k from 1 up to n in that order:

- Choose a subarray[†] of p of length exactly k, and subtract 1 from every element in that subarray.

After completing all n operations, your goal is to have all elements of the array equal to zero.

Determine whether it is possible to achieve this.

[□]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

[†]An array a is a subarray of an array b if a can be obtained from b by the deletion of several (possibly, zero or all) elements from the beginning and several (possibly, zero or all) elements from the end.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line contains the value n ($1 \le n \le 100$) — the length of the permutation.

The second line contains $p_1, p_2, \ldots p_n$ ($1 \le p_i \le n$) — the permutation itself.

Output

For each test case, output YES if it is possible to make all elements of the array p equal to 0 after performing all the operations; otherwise, output NO.

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

Input
4 4 1 3 4 2 5 1 5 2 4 3 5 2 4 5 3 1 3 3 1 2

Output
YES NO YES NO

Note

For the first test case, we can proceed as follows:

- $k = 1$: Choose the subarray [3, 3]. The array [1, 3, 4, 2] becomes [1, 3, 3, 2].
- $k = 2$: Choose the subarray [2, 3]. The array [1, 3, 3, 2] becomes [1, 2, 2, 2].
- $k = 3$: Choose the subarray [2, 4]. The array [1, 2, 2, 2] becomes [1, 1, 1, 1].
- $k = 4$: Choose the subarray [1, 4]. The array [1, 1, 1, 1] becomes [0, 0, 0, 0].

Thus, we have successfully reduced the entire array to zeros, so the answer is YES.

For the second test case, it can be shown that it is impossible to reduce all elements to zero.

For the third test case, the process is as follows:

[2, 4, 5, 3, 1] → [2, 4, 4, 3, 1] → [2, 3, 3, 3, 1] → [2, 2, 2, 2, 1] → [1, 1, 1, 1, 1] → [0, 0, 0, 0, 0].

The bolded values indicate the subarrays from which we subtract at each step.

For the fourth test case, it can also be proven that it is impossible to make all values zero.